ActInf Textbook Group ~ Cohort 5 ~ Session 3 (Chapter 1, Part 2) ~ 10/3/2023

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all right hello welcome everyone it's October 3rd 23 and we're in cohort 5 in our second discussion of chapter one we'll jump over to the uh cohort five questions it's not too late to add more just please feel free to copy in anything that you'd like us to explore but before we get to any of the questions is there just any kind of top of Mind piece about chapter one or some specific passage or some figure or something that anybody wants to share about chapter one before we jump into the question specifically okay okay let's go into the questions we we'll just go through them and again you can continue to add more or upvote them as as we continue question about modeling as action it seems that modeling is still abstract and based on observing and making effort to understand Dynamics Dimensions behaviors of given systems I.E modeling still seems theoretical possibly distinct from applying models and interacting with systems in the wild of course observing and documenting are also actions however the systems and behaviors being observed would be the primary actors expressing a rich complex range of acting in cohort 4 chapter 1 discussion arose a question about observing versus acting I guess it's well documented there inextricably intertwined what are anyone's thoughts on this for me um the perception action Loop seems to be more in line with you how we're making um policy decisions and not necessarily at this point about how we're mapping how we're moving um and so you I don't know if that's not you if if we can if we can kind of avoid conflating those two just to get the gist of where we're at in the in the book for me seems to be Cent thank you um Jesse I'm still on the phone for like the last day um yeah that that reminds me of like Maxwell's demon you know in a certain respect where even through making those choices you know it's it's an action so even the generating of the model and making that decision it's work that's also very um interesting point um it seems like the the questioner recognizes that um making a model is something that a person can do like a a researcher or scientist may do it so if you're thinking from the perspective of the modeler of course modeling is an action um applying a model that could be both abstracted and still an action I mean you could abstract away from the rat in the teamas like an example that will come back to or a frog jumping out of the hands which we'll come back to you can abstract that situation and model it and then you could apply that model but once you're in the map as opposed to the territory then there might be things that the territory could do like the rat could jump out of the maze but your model might not reflect that um and this is also a very rich question about observing versus acting we can revisit but action is not just overt gross bodily movement Act is conceived of very broadly uh in arbitrary spaces and so there's several ways that even in a situation that might be seen as a passive a passive observational setting there's several ways which action comes into play so visually icting is a a situation that's been modeled a lot in active inference so looking through a visual scene the is still an active policy selection ongoing about where the eyes move but that's still bodily in a sense but even more cognitively perception um is augmented by attention and attention in active inference is modeled by an internal policy decision so kind of like where to shine the internal Spotlight or how diffuse or how Focus to shine the attention Spotlight that's modeled as an internal or Co covert policy so there's definitely a lot of ways in which action and perception intertwine through through the loop and otherwise and um this is ironically or not a theoretical question about modeling being theoretical or not whereas once somebody puts something on paper and gets into the process of iterated modeling then it just becomes resolved in the actual what it is that they are doing and once you put the model on a piece of paper or computer it's not uh it still may be theoretical but but it it is also embodied so so just to clar to clarify something restraint would be considered an action is that correct restraint you're not taking action choosing to not take action that's a that's a great question there are several um ways in which uh timing your take is modeled often uh kind of like a null action is provided as an affordance so the rat is in the te-as they can go up down left right or stay and then Through Time it might select staying so it's like selecting an action but it's kind of like a null action so in one conversational sense you could say it didn't do anything but because the model must update every time step you have to provide an affordance that that does nothing so the selection is the action go ahead doesn't um a null action really make more sense there's there's two ways that null action can be it's not it's either not modeled or it's uh intentionally um nked and so once you get into nested models null action makes uh more sense as an intentional negation of action so where would indecision lie or would that just be yeah how would you frame indecision in that context O Okay so first to the um um let's say we're writing out a number so this is our our our act our action space writing out a number that's you know we're writing out our favorite number we expect and prefer it to be you know this number that we love and um this could be modeled as a single level model so then the affordances would be all the numbers that you could write down or it could be like a nested model where we're going to write down like the hundred's place and then the T's Place and the ones place and so like all the decisions you can make about the ones place are nested within the T's Place um so that's that's the kind of nested architectures or you could make it a single layer model to to um reference Andrew's point in decision um and then let's just say we include we have one model where you can write down zero through nine those are so every click of the model it must decide zero through nine another model you could say zero through n or pause and so that model depending on how the parameters are Set uh set so essentially how the um cognitive diversity is in that moment maybe one agent just Waits Waits Waits Waits Waits Waits Waits another agent writes down a three in the hundreds place and then Waits um indecision you could kind of un unpack that in a few different ways One agent might have very high Precision that they know that they want to wait so that would be indecisive in a way about what number it would be but they would be highly precise about what policy they wanted to engage in another flavor of indecisiveness would be if the um different options different affordances were kind of equally or similarly Salient so that kind of indecisiveness is like there's multiple options and I'm not sure which one to pick but the these are all these are great comments and and it just shows how even um behavioral settings that are like almost too simple to to believe like just writing down a single digit they they give us a handle into different cognitive phenomena memory attention preference decisiveness all these different things so that's why simple toy examples are used in the textbook because even very simple behavioral settings can give clear um it's like a example in a grammar book to just exhibit sentence structure these are like simple sentences of cognitive well-formed settings that help us practice fluency in the ACT INF anology Francis I was thinking about the uh way that acts and observing are not actually different and I think for me one of the differences between indecision and inaction is in actions kind of intentional maybe I'm waiting to learn something but when I'm truly stuck and uh this does happen to me uh in decision I feel almost like I'm not actually perceiving anything either I just um I'm in distraction mode I'm not actually picking up anything that's going to move me forward um yeah very interesting that's a um almost a uh analytical Auto ethnography or analytical Auto phenomenology like entering in at our own experience and then digging or trying to investigate like starting with what is it like to dot dot dot yes yeah and then like but what is it like to that's a very highlevel cognitive phenomena like a pendulum or um a simulated entity we don't need to appeal to it having like a higher order self-concept so um that's the double edge of entering in at the level of our own experience on one hand what else could be more tangible and immersive on the other hand the kind of phenomena that we might be dealing with are not necessarily like basil or the atomic or the essential phenomena but that's the fun is to then explore how how could how could we compose what it's like at a higher level from and and there doesn't need to be only one account and certainly not early in that research question so then just thinking divergently and building up different ways that it could be modeled understanding modeling as action like scientific modeling as real action in the world also it it just it opens many doors it just helps see um scientific pluralism in a new light like models are proposed maps of territories so building different Maps comparing different Maps that's model selection model comparison cool boxes aphorism uh all our models are wrong but some of them are useful exactly never been more true Kristoff yeah I was just kind kind of wondering like if um action in the sort of active inference setting can be really understood um much more technically in in the sort of you know thinking about this um in a sort of more like a graphical way in as there's some there's some sort of State machine that you know represent representation of the internal stat of an agent and whenever there is any kind of transition in that that's essentially an action something is happening and what is the meaning of that action is completely arbitrary we may um you know in our model we may assign the meaning of that action to be some kind of mental transition it could be essentially as we discussed the null action like I I do believe that you know not doing anything is very different from doing nothing doing nothing it's a very it's a very active State and it's not easy to be doing nothing consistently it's it requires a tremendous amount of practice um so but it's up to us to effectively assign the meanings to this but from the perspective of the formalism that really is kind of just a just just a state transition and there's some sort of you know effect that is being emitted by by the model by doing that state transition that is some some something can be observed um from the perspective of looking at the at the model as as it's kind of running so when we when we're updating to State absolutely two ways in which we'll see action as a variable in a graphical model or in a in a state machine here in more of the um particular partition representation style action is just as you said the the variable that's emitted by the cognitive entity of Interest so it's what goes back into just like perception it's not taking on some philosophical baggage of like being experienced um it's just the incoming statistical dependencies and in that view action is the outgoing statistical dependencies and then what we'll come to in chapter um four is action um here symbolized with pi for policy action is just what possible um affordance what what action selected intervenes how in the causal process of the world so yes it it it ends up being operationalized with a variable in a model once it comes to this stage you can also um generalize learning as action right so you can say action is extrinsically oriented or intrinsically oriented learning as as action let um I just searched for sanit Smith because um yes various mental or cognitive processes have been slash can be understood as action like if you have some belief distribution you could understand that movement as being a movement in a space and so hence an action in a space and um Chris fields and Mike Levan have talked about intelligence as competence in navigating arbitrary State spaces and so none of that requires overt bodily action this not the only kind of action being discussed and then once once you open up the door to thinking about internal action using control theory then that's been applied to attention that's been applied to learning as action and other cognitive phenomena okay awesome great question okay about our discussion on variational free energy as lower bound of a posterior distribution I found a good primer of ways of deriving evidence lower bound elbow is how some in in the basy in statistics and uh statistical physics world that's another like synonym for variational free energy um if anyone with a stronger computational bound would be willing to walk less proficient of us through the guide it'd be greatly appreciated I'll post a link in the notes if anyone in is interested in learning it comment I also posted in math learning group math learning groups just kind of um it's another math learning space people can and read through it Peru there's no like side group um there's also currently no other meeting time but in some Past Times people on the more uh basic or fundamental math learning side of the picture or more on the like rigor and and their improving side they've wanted to kind of specifically focus on the math so um perhaps comments on that question or or or um you know write more here or write a a comment or anything if you want to explore that more but looks good and and we'll come to some of these like mathematical technical things like they will come up in the coming chapters but it's great that that someone's thinking also ahead about this in chapter one cool well um if anybody has more questions they can just paste them in um this cohort 5 Section we can look to the overall questions on chapter one and kind of look at some of the most upvoted questions and just kind of review these questions that have been posed some of them are are are totally core great points or someone can just raise their hand and and like please feel free to just add any any point or ask ask any question um Andrew and then anyone else yeah I just thought of a question um I can add it too one thing I'm curious about the whole uh active inference Paradigm and it's Associated Concepts Vision uh mechanics and and whatnot is uh whether there's a good idea of what's really state-ofthe-art and sort of loosely held beliefs versus things that are really cementing and and um propagating uh as like pretty darn sure this is the way we should think about um various phenomena so what are yeah what are the loosely held beliefs in this space versus the strongly held ones great great question there's there's um there's a lot to say on that and um I I I hope as you know through time we we'll we'll continue to to develop better better answers to that [Music] um I'm thinking of how how to um reflect what has been happening very rapidly in the last few years to to address your question I'll just give kind of one historical note just to kind of plant the seed there um Carl friston and colleagues in the late 1900s working on neuroimaging and fmri in the neuroimaging unit at UCL and so that's the development of the statistical parametric mapping or SPM toolkit and so that's the most widely used neuroimaging Sensor Fusion toolkit that's what made Carl friston the most cited neuroscientist at the beginning of 2000 Carl friston and colleagues begins developing what was the free energy principle which was this idea that there could be a unified imperative or a unified principle for all that cognitive systems do unified imperative for perception and action and in some papers between 2006 to around 2012 there were a lot of um initial foras as far as I understand there's no major branch that was like totally repudiated however there were different equations where you know a Tilda had to be unpacked or a term had to be added um between around the the 2010 to decade there were increased applications which provided a different kind of support or belief confirmation which is to say it wasn't just about the analytical formalisms like transforming well between each other but also brought in like a level of explaining and predicting empirical data and uh for example the work of Ryan Smith and computational Psychiatry and during that decade also there was a cohort of um graduate students and and other researchers like Maxwell ramstead axle constant at all at all who were beginning to say well free energy principle for the brain which was like Carl friston's very well-sited 2010 paper it's like well it's not just about the brain we can expand this to think about like ecosystems and and so that kind of brought in like this multi scale modeling angle and then over the last few years um much of which you can find reflected in the live streams some of the specific angles that connected to physics which is now known as basian mechanics but basian mechanics was not a term used in this way several years ago but basian mechanics brought in a new le level of of rigor and grounding and it recontextualized some of those formalisms that had been proposed going back to 2006 and everything it recast those in terms of the um the generalization Direction in physics and put basian mechanics as a mechanics along which but not not a reductionistic or or um material account but a mechanics of basian systems and put that alongside Newtonian mechanics Thermo informational and quantum mechanics live stream 49 with Dalton Sak divid devel has some of that discussion and on the quantum side live stream 40 series with with Chris fields and also Chris fields's ongoing Physics course those represent some of the most ongoing developments in um re understanding free energy Principle as principle active inference as implementation approach and then basian mechanics as how that happens and again that didn't necessarily repudiate earlier work it helped understand its um scope though and then in more targeted areas people occasionally do write sort of review pieces which we can talk about but um you know the state of the field is is in some ways what people say here and in the literature that's assembled okay yeah so um to synthesize what you're saying um essentially the the newer research is uh reconceptualizing and clarifying earlier Stu um and so so helping put it on more solid epistemic foundations so so really the new stuff is pretty uh there's a lot of uh truth behind it it's not um sort of tentative exploration in terms of like the beian mechanics seems to really getting be getting getting a foothold in the physics world and um rather than being sort of tentative I would I would agree with that I think that's that's for each okay person to evaluate I'll just point to this 2010 F the 2010 first in paper that I um mentioned there this is like a very provocative early positioning with like the free energy principle and then understanding how out of the free energy principle potentially various special case cognitive phenomena but still vast and Broad phenomena like model selection and evolution could be described um as a special case of free energy principle um in the setting of The Action perception Loop now there's been subsequent developments on like what do we mean by the action perception Loop um what about the arrow that can that can plausibly connect you know across the blanket like so some details have been coming into better focus but but but also it's only been like months to low numbers of years that tends to potentially only small orders of magnitude more numbers of people having even been able to reckon with the quantum formalism or basy mechanics or G Theory so ongoing Susan um to follow up on Andrew's question yeah I I'll start with uh reading kind of the summary of chapter two which I thought we were going to be going every chapter two but um but it it offers kind of a generalized um theory of active inference is a theory of how living artifacts and I you I I kind of put there organisms underwrite their existence by minimizing surprise variational free energy via perception and action and what I take away um from you know the more I read about it which I which I like because so much of psychology tends to be you know in decision management it tends to be prescriptive but taken out of context it's more wrong than right so what I kind of like about the um what I'm reading is that it's not trying to be prescriptives is trying to give us a framework to uh to explore how individuals can build models and how we can build models that are more closely resembles reality so for what that's worth that's a great Point um thank you raore I I love this historical element and I'll just point to one more um kind of piece of recent history um in 2019 Carl friston wrote this long monograph a free energy principle for a particular physics particular is a little bit of a pun double on Tandra it's a specific physics like if you want to make a different physics you can people there's Newtonian physics there's relativistic there's all these different physics and so this is just one specific physics so if you have some qual with it you can version and modify it or you can develop a different physics but also it's not just a specific physics or a peculiar one it's about a particle it's about the cognitive particle or just like the generalized particle and so it was this paper in 2019 that really signaled an inflection point with like we're going to be talking about the particle which is like the blanket and the internal States that's going to be like figure and it's going to be partitioned off from ground or the niche or the environments so it's like if that partition can't be made between the system of Interest blanket and internal States the particular States here if that can't be partitioned off from the niche you don't have a noun or a thing to point at that's like almost tological but like but but um if you can make a distinction there's a difference and so that's a thing and that's going to be the particular States and then there's some some subpartitioning of of certain um pieces that we might call our attention to or develop formalisms around but it was in 2019 and then it was a reading group with Maxwell and others that had that brought a a relatively lot of attention to um the physics here and then that um that was kind of incubated in the 2019 era led to Dalton and others really um strengthening the physics side and connections Kristoff um you know I think there's a interesting development happening right now in the uh in that space you know like you mentioned that uh if you can't draw a boundary around the thing if you can't if you can't make that blanket then you don't really have a thing and I think some some work that is yet to be published that Maxwell ramstead uh mentioned a couple times on various podcasts recently they're formalizing the notion of essentially a fuzzy blanket where there is some kind of like index into the blanket blanket n of of of the blanket like how how how fuzzy it really is and uh you know my uh my prediction is that it's going to it's going to uh turn out to be something along the lines of that that that the distinction is always somewhat arbitrary that at a certain scale you have a thing in a certain scale there is no thing you know almost like the the Buddhist concept of no self is going to show up in the mathematics of it that depending how you look at the system the internal States really are a thing and and and and from from a certain perspective and from a different perspective that thing just simply disappears if you if you start really looking for it it's like when you start start to look for the boundary of the um I don't know a cell you're going to find that that boundary is becoming fuzzier and fuzzier the closer you look right uh the the the more Jagged and fractal and and and harder to Define what what exactly is the internal state from from the from from the external state so um it's a it's an interesting it's very interesting to see how that stuff developed you know like when in the I think the there were a lot of ideas how to draw that boundary very it was just kind of like a hard boundary that you have to have this distinction uh that has to be very uh very strong between the internal States and external States and now it's slowly dissolving as the mathematics develops and I think this reflects the fact that mathematic is ultimately mathematics is ultimately an intuitive discipline um you know this principle the principle is true and you can you can sort of see it for yourself but actually properly formalizing the mathematics in such a way that it's actually airtight in in all cases that always takes a long long time you know calculus was full of holes for I don't know like a long long time like probably until uh um uh until um Whitehead and uh what's this the I forgot the I forgot the other guy princip of Mathematica came along and started formalizing things and and so and so on and I think the same process kind of has has been happening in the um in in the free energy principle space as well that a lot of these ideas started started off a little bit um you know based on certain mathematical assumptions that now have been revised like er ergodicity used to be somewhere there and I think now they they they they refor reformulated some of the ideas without it um and it's becoming Much More Much More airtight and that's just a natural progression great points thank you Kristoff yeah just a few Point uh pie on that um here's a 2022 uh paper um perhaps not the the only or the total work but the blanket index was proposed which is kind of like a zero to one Continuum about like fuzzy or dynamical or wandering blankets but being able to um recognize from empirical data and not have the particular partition be in a priori that then dictates the subsequent analysis but enable the discovery of things to to be um part of of the analysis process that's those are great and then also um especially with Chris fields's work on the quantum information sciences and Mike Len's work on poly Computing there is also the um increased focused on like The Observer playing a fundamental role like you can't take a view from nowhere on what computation or what cognition something is performing if it's encrypted to you it's going to look like noise if it's not encrypted you might understand what it is and then exactly like what you said about the spatial and temporal scale like if your time scale is uh 100 years then an eoli lives and dies as as a statistical noise and you you would not be able to perceive that as a thing whereas if it was um you know a Pico second time interval you wouldn't perceive action occurring and so 100% it's about like zooming in zooming out and finding Salient spatial and temporal scales of analysis which comes back to the modeling as action there's just no model without a modeler and there's no observed without the Observer and and so some of these points are and and and then you pointed to to some more technical discussions on ergodicity and and requirements around that but these um um we're reading the first textbook of active inference which realistically is not so much longer than for example step by step it's it is different but um this is not a massive textbook however what sanj namjoshi is doing is incredibly comprehensive and may um his textbook coming out within the coming you know 12 to 18 months or whatever it is that will speak to this question in a way that that the textbook that we're discussing here doesn't um lay it on so thick but um yeah these are these are epic discussions and and this is just like being in the game to even enter the slipstream to try to understand these and also like to understand where it's like okay erodic it's just like all right it is that that is how it is and people are working on those questions um Susan yeah it's definitely upending um a lot of of Economic and other affable um models that uh just don't suit the world anymore um I'm curious um to explore uh with what you mean um kristo am I saying that correctly Kristoff when you're when you're talking about airtight so when you talk about airtight in mathematics can you expand on that just a slight just a bit um so let me let me give you an example um from mathematics that's kind of vivid from my education the first time I encountered uh for your analysis you know it's like it's it's a beautiful piece of mathematics but there's a problem with it it just does not converge at the ends of the intervals in in in a lot of really really simple cases and it took quite a lot of work to actually make that work mathematically right you have to um get into schwarz's family of rapidly decreasing functions and um generalize notion of functions distributions and so on so forth and and bring bringing that in incredible mathematical rigor um you can actually make um for air transform air tight so to speak that you can finally really uh stop making the handwavy argument that oh you know like I have this limit of some kind of gaussian um gaussian function that in the limit actually approaches the direct Delta but there is there's a precise way of defining that in terms of sches rapidly decreasing functions and so on so forth um and then you actually get a formalism which you act you you get proper results all the time but it never really stopped anybody from making tremendous use of um f Transformers it was right like with these little holes which just knew they were there um Engineers would be aware of them and that that that that never really stopped anybody uh it's it's validity you know the the validity of the idea was not um was not questioned just because there were certain cases in which uh it would break down and especially that intuition Behind these these sort of limits and calculus is another good example I think um you know Russell and Whitehead they they got to um formalizing mathematics precisely because they were worried about this you know this this notion of a limit in Calculus if I if I remember my history correctly uh what what do you mean when you know something this this um something tends to be it tends tends to become infinitely small what what what does it actually mean precisely like it it's always been kind of like a hwa handy argument um but it never stopped mathematics from just working right like there is a there is a certain um geometry to it that you can just simply see and symbolically in in some sense non-constructive mathematics is is always like that unless you have a completely constructive system computational system effectively and and you derive something from from an axotic Foundation then your your mathematics is is is quite likely going to be full of full of some holes and you can you you can get into some really interesting philosophical arguments like what why is it even possible that we can do something like that to without having this precise computation to jump over the threshold and say like well you know I kind of I have this intuition that this is actually going to work out and it does and it's it's a it's a fascinating question but that's uh I think this is the kind of process that we're what we're seeing is this kind of thing where we started off with intuitive ideas that came from a lot of um experience of applying mathematics in in in in in in in this kind of setting um then various attempts to formalize that using the mathematical tools that were available um at the time finding certain cases where these tools were insufficient and you know rederiving this for this this formalism again and again until until it becomes um uh you know uh again like I'll use the same term or tip I hope I hope that makes sense yeah absolutely actually I'm very sensitive to what Andrew brought up in terms of you know what is the underlying beliefs I I actually you know constantly um I'm trying to understand ideologies because they're they're underneath beliefs and in fact one of the statements I wrote down from chapter two was which was you know an assumption was whether we adjust our beliefs or our data depends on the confidence with which we hold the beliefs I I think that kind of speaks to what you're saying in terms of confidence levels um and um you know and and so you know when we kind of unpack the you the surprise the variational surprise and consider that that is really about uncertainty if we can you build that mathematical model that um that more accurately um aligns with reality then um whether that be on an individual basis or group basis or basis doesn't matter I mean we're going to start getting better results so so yeah that's that's a great uh explanation of math being a non-math person thank you yeah th this was great I I love this point like like we might be able to have an intuition like okay what if you take a a gaussian so like a a bell curb distribution and you just make it sharper and sharper and sharper and sharper it's like can we just say that it's a spike it's like well maybe not necessarily but in a given application that might work perfectly well and then those like sort of intuition looking across the Grand Canyon you know could we jump across it could we build a bridge or is this like an incross Gap th those are the research um adjusting our belief or the the data this is in reference to the two ways that free energy can be minimized so the two ways in which surprisal can be bounded which is change the world through action or change the Mind through updating the cognitive model which might be considered learning or perception or attention depending on on which way so here's sensory data coming in we don't want to have our thumb on the sensory scale like that's kind of like a trivial and ultimately um less effective way to let's just say um weigh the amount that we want to weigh well yeah there's a trivial way to make that happen which is just change the number on the scale so we take the thumb off the scale sensory data we don't directly control what are the two ways that the particular State can then come into uh come to bound its surprise or minimize its surprise change your mind change the internal state or change the world through action those are called the autonomous States just the internal and the active States so that's an example of like a subset of the particular States it's two out of the three particular States but then we can talk about particular states to differentiate the thing from its Niche so that's these three and then you might want to talk about just the blanket States just the the two in the middle perception in action or you might want to talk about the autonomous States just action and internal States because those are the controllable ones in an optimal setting if you can control what to think and control it to do and then your actions through the niche end up influencing your observations and hopefully linning them with your preferences that's pragmatic value cool and figure uh one and two figure one one kind of grounds Us in fundamental agent World interaction so this is even simpler than the four-fold partition that we see here this is just kind of the two fold partition just figure from ground or like agent from Niche agent-based modeling and then figure one two lays out quite literally like the road map on how we're going to get to active inference in the coming chapters the low road is a how question it starts with basian Statistics and it's going to work up towards basian cognitive model models but basian cognitive models could be of any kind you could interpret a program as a basian cognitive model or there's nothing about an arbitrarily constructed basian model that makes it like adaptive in the niche but that's the how and then coming from the high road which is chapter three we have this persistance or repeated measurements imperative things that can't be repeatedly measured Meed are going to be understood to be noise and not signal they're just not persistent long enough to be measured so as an observer it isn't a thing in our data if it's changing at a time scale faster than our um like persistence can understand or slower um and then when that gets turned back on the agent itself as long as you're remeasuring your body temperature to be in a homeostatic range you're alive and so that's how the um survival or self-organization or self- evidencing is understood but this doesn't uniquely suggest a how there might be different ways to develop self-organized self- evidencing surprise minimizing marov blanketed systems but the RO map of this textbook is that in Chapter 2 we're going to learn about the how and about how basian statistics and updating can be used to develop models of perception and action and then you know refresh how systems can be understood as embodying this principle of persistence because we're talking about systems that do exist even if they're only proposed to exist and then those two roads come together in active inference and then right in chapter four we're going to be talking about the generative models of active inference which use this how for why and there they are um yeah in the last minutes anyone have any other thoughts on one or about like where we're heading in the coming weeks I do have one question about um kind of tying into my earlier comment on beijan mechanics do you have a feel of what um whether we touch on the connection mentioned in basan mechanics between um I I believe it's called entropy maximization as like the external View and I think the textbook is mostly this sort of internal view but whether we touch on that other piece at all great comment so um the work of Dalton and Maxwell and others um in check out live stream 49 um or the guest stream with uh with Dalton let's see what number that was guest stream 23 rebooting the free energy principal literature in this work as you alluded to it came to be understood and demonstrated that the free energy principle is like the view from the outside on persistence and then the constrained maximum entropy principle CME is like the view from the obs Observer which is like we should kind of be like maximally agnostic about what that thing is going to do but but no more agnostic than that and then um the view from the inside um is the free energy principle and they unpack this in work um it's not heavily addressed in this textbook as it was more of a recently breaking and more technical point but these discussions help connect Fe with CME cool well um people can can make any final comments otherwise this was an awesome discussion on one and then in the coming two weeks we'll be heading into the low road with chapter too so in the last minutes does anyone have any other comments that they want to make just about heading out of chapter one or heading into chapter two uh I have one like there there's something interesting that I run into recently that I I think to me it's it's very much related to active inference uh but it's um it's it's it's a bit of a sort of a uh why they suppose relationship I I recently run into a bunch of interesting literature on the subject of um inside meditation so um essentially Buddhist Traditions from the terada tradition or more recent stuff like mahasi sa's um Manual of inside and so on and ultimately uh you know everything that that that Gama Buddha uh was was was trying to teach people and like all these all these Med all all these meditation teachers uh that um came after him are are trying to teach is effectively a method for um building a generative model of the Mind from the inside and it's extraordinarily simple it it really is about paying attention to what's happening in the mind and then having a vocabulary to label what's what's going on I am thinking I am intending I am looking I am moving I am feeling this I am I am I am thirsty I am cold I am you know uh feeling pressure or uh pain or or pleasure and so on so forth and the entire um system is based around literally noticing that this is what's happening and constructing ever so ever more sophisticated models of what's actually happening with the mind um and interestingly like the only the only thing that there is for a person to do um to acquire all these skills is just observation because the the basic premise is that there is no there is nothing you can do to make your mind do anything it will do things on its own and all you can do is to kind of notice that this is what has happened and then hold an intention that this is what you would like to happen more and if you hold that intention and you sort of project positive energy towards the outcomes you want over time uh this reinforces the the the the the the um you know occurrence of these events uh more often so you become more mindful the mental states that you want to cultivate increase the mental states that you um that are UNH wholesome diminish and and so on so forth but it's really fascinating if you start thinking about this in terms of literally constructing a generative model that I have a model of what my mind might be doing and in every moment of perception when something is happening it hits the model and I can and I can see like oh these are you know of all this of all these internal of all these states that I have mapped out this is um this is this is what's happening here so this relationship between you know the mathematical Theory and something that you can actually phenomenologically investigate by yourself by you know just sitting quietly and and and looking at what what your mind is doing is is really fascinating thank you we're at the end of our time so I will close it but awesome discussion thank you cohort 5 and looking forward to seeing you in the coming weeks and please add questions add writings you know if this Sparks something you just just add questions and and insights and we will um go from there so thank you and farewell see you everybody bye